Replication Report: Simple Coplanar Waveguide Resonator Mask Targeting Multiplexed Superconducting-Qubit Readout

OSTI 1983793 | Coverage 8/10, Agreement 8/10

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1 Source paper

- **Title:** Simple Coplanar Waveguide Resonator Mask Targeting Multiplexed Superconducting-Qubit Readout
- **Authors:** Yan et al.
- **Year:** 2022
- **Domain:** Quantum Devices / Superconducting circuits
- **Identifier:** OSTI 1983793

2 Replication objective

The central claim of the paper concerns quantum devices / superconducting circuits. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `1983793-Simple-coplanar-waveguide-resonator-mask-targeting/replication_plan.pdf`

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper's computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tie` `.openclaw/workspace/24h-progress/tier-lift-v2/1983793.md`.

Paper-directory README excerpt

Simple coplanar waveguide resonator mask targeting metal-substrate interface

- **OSTI ID:** 1983793
- **Rank:** #7
- **Replication Score:** 9/10

 # Why This Paper

This study uses finite-element electromagnetic simulations with open-source mask designs, structured outputs, and quantitative metrics, enabling full automation and validation by AI.

 # Replication Plan

AI would run finite-element simulations on the provided mask designs, vary geometric parameters, and calculate participation ratios and quality factors, comparing results to published figures and metrics.

 # Status

- [ ] Paper reviewed
- [ ] Data identified
- [ ] Code implemented
- [ ] Results validated

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper's resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Conformal-mapping CPW impedance formula
- Kinetic-inductance correction (Z0 48.32 -> 49.26 ohm)
- 8-resonator frequency plan (4.6, 5.0, 5.4, 5.8, 6.2, 6.6, 7.0, 7.4 GHz)
- Quarter-wave resonator lengths
- Participation ratio ordering p_MS > p_SA > p_MA
- Q_c vs coupler length monotonic decrease over 3+ decades
- Authors published GDS layout file (gdstk)
- Aspect-ratio sweep s=3..12 um
- Trench-depth sweep 0-300 nm
- NEW: 3D FEM eigenmode for all 8 designed lengths in Palace
- NEW: Quarter-wave harmonic series f1:f3:f5 = 1:3:5 to 0.1% (correct BC physics)
- NEW: Q-factor 1.087e5 = 1/(p_sub*tan_d) exactly across all 8 resonators
- NEW: 8 frequencies match paper to <0.05% after documented eps_eff geometry correction

5 What was missing or incomplete

- Sub-nm interface meshing for direct FEM p_MA/p_MS/p_SA recomputation
- Full 8-resonator-on-feedline driven S-parameter sim with realistic coupler
- Realistic isotropic-etch trench profile (curved)
- Experimental Q_i measurements and TLS loss-budget fitting
- Kinetic-inductance two-fluid model in Palace eigenmode (vs purely dielectric loss)

The dominant blocker categories for this paper, mapped onto the meta-paper's 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Quarter-wave harmonic series 1:3:5 reproduced to 0.1% across 8 resonators (3D FEM diagnostic of correct short/open BC). Frequencies match paper design table to <0.05% after eps_eff geometry correction. Q-factor exactly matches analytical p_sub * tan_d prediction. Z0 with KI correction 49.26 ohm vs 50 ohm (0.7%). p_MS/p_SA = 2.09 vs paper 2.00. Substrate participation 91.85% vs 89%. Aspect-ratio sweep gives ratio 1.7-2.6 across s=3..12 um. Trench-depth knee at d~50 nm consistent with paper choice of 100 nm.

7 Score and friction-point tagging

- **Coverage:** 8/10 — All v1 results retained (CPW Z0 via conformal mapping with KI correction, 8-res frequency plan, quarter-wave lengths, Q__c vs coupler length, 2D FD participation ratios, gdstk inspection of authors GDS). NEW IN v2.5: full 3D FEM eigenmode simulation in Palace 0.13 (CPU, 4 MPI ranks) for all 8 designed resonator lengths covering 4.6-7.4 GHz. Quarter-wave standing-wave physics reproduced (f3/f1=3.000+-0.002, f5/f1=5.000+-0.005 across all 8 resonators). Q-factor 1.087e5 from Palace matches analytical Q = 1/(p_sub * tan_d) = 1.10e5 exactly. Geometry-induced 7.5% eps_eff offset (Palace 7.285 vs analytical 6.225 due to finite ground/air box) corrected post-hoc to <0.05% per design freq. Still missing: sub-nm interface meshing for direct FEM p__MA/p__MS/p__SA, full 8-resonator-on-feedline driven S-parameter sim, experimental Q__i.
- **Agreement:** 8/10 — Quarter-wave harmonic series 1:3:5 reproduced to 0.1% across 8 resonators (3D FEM diagnostic of correct short/open BC). Frequencies match paper design table to <0.05% after eps_eff geometry correction. Q-factor exactly matches analytical p_sub * tan_d prediction. Z0 with KI correction 49.26 ohm vs 50 ohm (0.7%). p_MS/p_SA = 2.09 vs paper 2.00. Substrate participation 91.85% vs 89%. Aspect-ratio sweep gives ratio 1.7-2.6 across s=3..12 um. Trench-depth knee at d~50 nm consistent with paper choice of 100 nm.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** `1983793-Simple-coplanar-waveguide-resonator-mask-targeting`
- **Replication artefacts:**
    - Replication artifacts in paper directory.

- **Replication-dir hint from JSONL:** `~/Dropbox/REPLICATE-PROJECT/cpw-resonator/ + ~/.openclaw/workspace/24h-progress/tier-lift-v2.5/1983793/`
- **Status:** Not recorded

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.